Publicly Available Datasets — ML Fault Detection & Predictive Maintenance

All datasets listed below are free for academic research use. Direct download links provided. Colab-compatible via **wget** or Kaggle API.

Dataset	Source / Org.	Signal type	Task	Fault types	Size	Download link	Notes
CWRU Bearing Dataset	Case Western Reserve Univ.	Vibration (accel.)	Classification	Ball, inner/outer race — 4 severities	~500 MB	engineering.case.edu/bearingdatacenter	Industry standard; most cited; 48 kHz; 4 load conditions
NASA FEMTO / PRONOSTIA (PHM 2012)	FEMTO-ST / IEEE PHM	Vibration + Temp.	RUL	Run-to-failure, 17 bearings	~230 MB	kaggle.com (bearing-dataset)	3 operating conditions; standard RUL benchmark
IMS Bearing (UoC)	Univ. of Cincinnati / NASA	Vibration	RUL + Anomaly	4-bearing rig; run-to-failure	~6 GB	kaggle.com (IMS bearing)	Long degradation traces; good for temporal modelling
MFPT Bearing Dataset	Machinery Failure Prevention Technology	Vibration (accel.)	Classification	Outer race, 3 load conditions	~20 MB	mfpt.org/fault-data-sets	Small, clean; good for quick prototyping
Paderborn Bearing (KAT)	Paderborn University	Vibration + Current	Classification	Real + artificial damage; multiple types	~26 GB	mb.uni-paderborn.de/kat	Multi-sensor; ideal for motor current analysis (MCSA)
XJTU-SY Bearing Dataset	Xi'an Jiaotong University	Vibration	RUL	15 bearings run-to-failure; 2 axes	~1.5 GB	biaowang.tech/xjtu-sy-bearing-datasets	2 accelerometers per bearing; 3 operating conditions
SEUC Gearbox Dataset	Southeast University China	Vibration + current	Classification	Gear tooth faults, bearing faults	~4 GB	github.com/cathysiyu/Mechanical-datasets	Combined gearbox + bearing; two-stage rig
Ottawa Bearing Dataset	University of Ottawa	Vibration	RUL + Classification	Outer race, run-to-failure; varying speed	~2 GB	kaggle.com (Ottawa bearing)	Variable speed conditions; useful for speed-varying scenarios
PHM 2009 Gearbox Challenge	IEEE PHM Society	Vibration (multi-sensor)	Classification	Gear faults, bearing faults, shaft faults	~300 MB	kaggle.com (PHM09 gearbox)	Challenge dataset; labelled train + test splits provided
MaFaulDa (Machinery Fault Database)	UFRJ / PUC-Rio, Brazil	Vibration + current + tach	Classification	Imbalance, misalignment, bearing, bowed shaft	~6 GB	smt.ufrj.br/mfs	Broad fault taxonomy; tachometer included for RPM sync
FEMTO Degradation (IEEE PHM 2014)	FEMTO-ST Institute	Vibration + Temp.	RUL	Accelerated degradation; multiple speeds	~180 MB	kaggle.com (FEMTO 2014)	Extension of PHM 2012; additional operating speeds
CWRU + DIRG Gearbox	Politecnico di Torino (DIRG)	Vibration	Classification	Gear pitting, tooth breakage,	~800	github.com/DIRG-POLITO/Gearbox-Fault-Dataset	Open GitHub release; good for gear-specific research

Task legend: Classification | RUL (Remaining Useful Life) | RUL + Anomaly — All datasets are free, open-access, and loadable in Google Colab via wget or Kaggle API.